

# Clustering and Association Rule Analysis of Player Performance Data

This analysis identifies player profiles using clustering techniques and explores relationships between performance metrics using association rule analysis

PC1- overall player contribution , PC2- Distinguishes between defensive and attacking roles, PC3- playing style

## K-means Clustering

- K-means applied to principal component scores, leading to more meaningful results by reducing noise
- Number of clusters (K=4) selected using the elbow method
- Clusters differentiate players by role and overall contribution
- Cluster 1 represents low involvement players
- Cluster 2 indicates versatile or midfield players
- Cluster 0 and cluster 3 respectively represent low contributing players with defensive characteristics and highly contributing players with mixed role characteristics.

## Hierarchical Clustering

- Dendrogram shows increase in linkage distance at higher levels which was used to confirm the presence of around 4 clusters
- Dendrogram used to see cluster structure without pre defining the number of clusters
- Players merging at lower linkage distances show similar performance profiles
- Suggests potential for identifying direct substitutes for recruitment
- Higher linkage distances show more distinct player roles
- Cluster structure shows role specialisation within the dataset

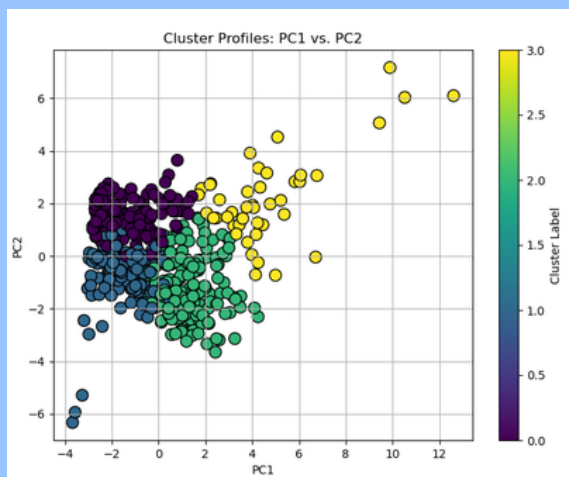


Figure 1: K-means clustering  
Clusters separate players by role and overall contribution

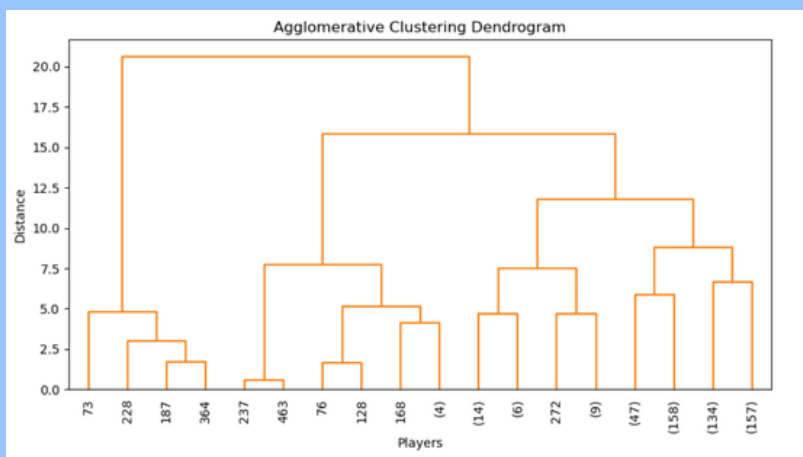


Figure 2: Hierarchical Clustering Dendrogram

## Method Comparison & Preferred Approach

The Rand Index (0.4199) and Adjusted Rand Index (0.1008) indicate moderate agreement between methods with the agreement being slightly better than random clustering. Increasing the number of clusters improves agreement but reduces interpretability. Furthermore, k-means is preferred due to clearer and more interpretable clusters.

## Association Rule Analysis

Rule 1: High Tackles → High Intercepts

Support: 0.3562 Confidence: 0.7138 Lift: 1.4463

- Strong and relatively common defensive pattern
- Players who tackle a lot tend to make more interceptions, showing consistent defensive involvement
- Defensive actions often occur together not separately
- Reflects typical behaviour of defensive players

Rule 2: High Assists & High Passes → High Goals

Support: 0.1725 Confidence: 0.5254 Lift: 1.7812

- Less frequent but stronger association
  - Goals are linked to overall involvement rather than isolated factors
  - Reflects attacking players who contribute in multiple ways
- min. support threshold: 0.02, lift threshold: 1.1 ensured rules were meaningful and not driven by random associations

## Evaluation

- Variation in playing style captured by PC3 is not visible in Figure 1, hierarchical clustering lacks clear centroid and binary transformation can lead to information loss. Therefore, some interpretations can be limited and some differences between players may be hidden, reducing reliability.

## Summary

The analysis identifies distinctive player roles and identifies players with specific performance metrics. The findings are useful for recruitment and squad planning, although the results are sensitive to the choice of methods and should be interpreted with caution.

Appendix A: Graphs and Tables

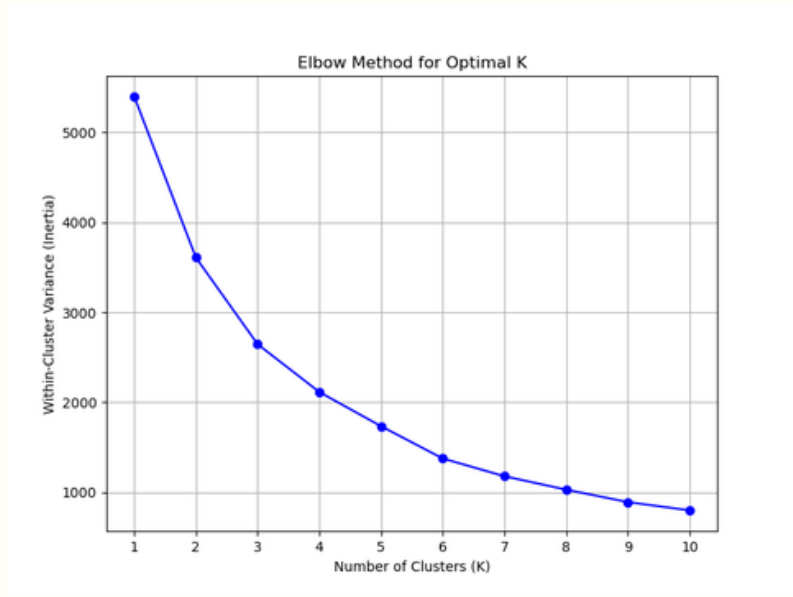


Figure 3: Elbow Plot

| antecedents       | consequents       | antecedent support | consequent support | support  | confidence | lift     | representativity | leverage | conviction | zhangs_metric | jaccard  | certainty | kulczynski |
|-------------------|-------------------|--------------------|--------------------|----------|------------|----------|------------------|----------|------------|---------------|----------|-----------|------------|
| (High Goals)      | (High Assists)    | 0.294991           | 0.456401           | 0.230056 | 0.779874   | 1.708749 | 1.0              | 0.095422 | 2.469494   | 0.588328      | 0.441281 | 0.595059  | 0.641970   |
| (High Assists)    | (High Goals)      | 0.456401           | 0.294991           | 0.230056 | 0.504065   | 1.708749 | 1.0              | 0.095422 | 1.421576   | 0.763019      | 0.441281 | 0.296555  | 0.641970   |
| (High Assists)    | (High Intercepts) | 0.456401           | 0.493506           | 0.261596 | 0.573171   | 1.161425 | 1.0              | 0.036359 | 1.186642   | 0.255682      | 0.380054 | 0.157286  | 0.551623   |
| (High Intercepts) | (High Assists)    | 0.493506           | 0.456401           | 0.261596 | 0.530075   | 1.161425 | 1.0              | 0.036359 | 1.156779   | 0.274414      | 0.380054 | 0.135531  | 0.551623   |
| (High Assists)    | (High Passes)     | 0.456401           | 0.499072           | 0.328386 | 0.719512   | 1.441699 | 1.0              | 0.100609 | 1.785916   | 0.563603      | 0.523669 | 0.440063  | 0.688752   |
| (High Passes)     | (High Assists)    | 0.499072           | 0.456401           | 0.328386 | 0.657993   | 1.441699 | 1.0              | 0.100609 | 1.589437   | 0.611613      | 0.523669 | 0.370846  | 0.688752   |
| (High Goals)      | (High Passes)     | 0.294991           | 0.499072           | 0.209647 | 0.710692   | 1.424026 | 1.0              | 0.062426 | 1.731467   | 0.422357      | 0.358730 | 0.422455  | 0.565383   |
| (High Passes)     | (High Goals)      | 0.499072           | 0.294991           | 0.209647 | 0.420074   | 1.424026 | 1.0              | 0.062426 | 1.215689   | 0.594428      | 0.358730 | 0.177421  | 0.565383   |
| (High Passes)     | (High Intercepts) | 0.499072           | 0.493506           | 0.324675 | 0.650558   | 1.318235 | 1.0              | 0.078380 | 1.449434   | 0.481926      | 0.486111 | 0.310075  | 0.654226   |
| (High Intercepts) | (High Passes)     | 0.493506           | 0.499072           | 0.324675 | 0.657895   | 1.318235 | 1.0              | 0.078380 | 1.464250   | 0.476630      | 0.486111 | 0.317057  | 0.654226   |
| (High Tackles)    | (High Intercepts) | 0.499072           | 0.493506           | 0.356215 | 0.713755   | 1.446292 | 1.0              | 0.109920 | 1.769438   | 0.616011      | 0.559767 | 0.434849  | 0.717780   |

Figure 4: Association Rules

## ***Appendix B: Cluster Summary***

| <b>Cluster</b> | <b>PC profile</b>             | <b>Description</b>                                                               |
|----------------|-------------------------------|----------------------------------------------------------------------------------|
| 0              | Low PC1,<br>Moderate/High PC2 | Low contributing<br>players with defensive<br>characteristics                    |
| 1              | Low PC1, Low PC2              | Low involvement<br>players with limited<br>overall contribution                  |
| 2              | Moderate PC1, Low<br>PC2      | Versatile profiles<br>(midfielders)                                              |
| 3              | High PC1, PC2 Varies          | Highly contributing<br>players across both<br>defensive and<br>attacking actions |

Table 1: Cluster Summary